

ELODIE A. HARBOURNE

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RESEARCH EXPERIENCE

DPhil (PhD) Inorganic Chemistry, University of Oxford (Hertford College) (2022–2026)

Proposed thesis title: “Understanding local structure using total scattering and Monte Carlo techniques”

Supervisors: Professor Andrew L. Goodwin FRS (Department of Chemistry, University of Oxford) and Professor David A. Keen (ISIS Neutron and Muon Source, UK)

Overview: I am interested in disordered functional materials, particularly where disorder is not random but correlated. Prussian blue analogues (PBAs) are an important and compositionally diverse class of hybrid framework materials with applications spanning a variety of different fields, including potassium-ion cathode materials and gas storage. PBAs can contain a wide variety of different distortions and defect structures, including correlated octahedral tilts, Jahn–Teller distortions, and hexacyanometallate vacancies. Gaining a better understanding of the interplay of the different distortion mechanisms is therefore important when seeking to optimise the performance of functional PBA materials. I use a combination of total scattering/pair distribution function measurements and Monte Carlo simulations to understand the local structure of PBAs and related perovskite materials.

Publications:

- A. F. Sapnik, P. A. Chater, D. S. Keeble, J. S. O. Evans, F. Bertolotti, A. Guagliardi, L. J. Støkler, **E. A. Harbourne**, *et al*,
- “High-quality ultra-fast total scattering and pair distribution function data using an X-ray free electron laser”, *IUCrJ.* **2025**, 12, 531.
- B. R. M. Tragheim, **E. A. Harbourne**, C. Ritter, A. L. Goodwin, M. S. Senn, “The interplay between Jahn–Teller distortions and structural degrees of freedom in pseudocubic states in manganite perovskites”, *Phys. Rev. B* **2025**, 112, 115119.
- E. A. Harbourne**, J. Cattermull, H. L. B. Boström, R. Witte, N. Roth, M. Pasta, D. A. Keen, A. L. Goodwin, “Rules governing Jahn–Teller order in Prussian blue analogues”, arXiv:2408.13169, submitted 23/08/**2024**, in reviews process at *Nature Communications*.
- E. A. Harbourne**, H. Barker, Q. Guérault, J. Cattermull, L. A. V. Nagle-Cocco, N. Roth, J. S. O. Evans, D. A. Keen, A. L. Goodwin, “Local Structure and Dynamics in $\text{MPt}(\text{CN})_6$ Prussian Blue Analogues”, *Chem. Mater.* **2024**, 36, 5796.

Conferences attended:

Posters: PCG Winter Crystallography Meeting (Oxford, UK), 2023–2025; 4th International Symposium on Negative Thermal Expansion and Related Materials (Padova, Italy), 2023, **1st Place Poster Prize**; 26th Congress and General Assembly of the International Union of Crystallography (Melbourne, Australia), 2023, **top poster research presented in the field of Powder Diffraction**; RSC Solid State Chemistry Group Meeting (Edinburgh, UK), 2023; Correlated Disorder (Aarau, Switzerland), 2024; Oxford Women in Chemistry Day (Oxford, UK), 2024, **Inorganic Poster Prize**.

Talks: RMC-8(+1) Reverse Monte Carlo (Budapest, Hungary), 2023; ISIS Crystallography Group Meeting (Oxford, UK), 2024; 5th International Symposium on Negative Thermal Expansion and Related Materials (Porto, Portugal), 2025; 35th European Crystallographic Meeting (Poznań, Poland), 2025; Conference on Analysis of Diffraction Data in Real Space (ESRF/ILL Grenoble, France), 2026.

MChem Chemistry (1st Class), University of Oxford (St Hugh's College) (2018–2022)

Thesis title: “Cooperative Jahn–Teller order in Prussian blue analogues” (1st Class)

Supervisor: Professor Andrew L. Goodwin (Department of Chemistry, University of Oxford)

Part IB 2021 (3rd Year exams): 1st Class

Part IA 2020 (2nd Year exams, cancelled due to COVID-19): 1st Class (92), with highest marks in the year for Inorganic Chemistry (96) and Organic Chemistry (94).

AWARDS

- Lois Vernon Prize for the St Hugh's College chemist scoring the highest practical mark (80) after Part IB.
- Recipient of a scholarship awarded by St Hugh's College, University of Oxford, for consistently scoring first class marks during second year and third year.

RESEARCH SKILLS

- Synthesis of powder samples of Prussian blue analogues
- Coding – Fortran, Python, MATLAB, Wolfram Mathematica
- Monte Carlo simulations
- Total scattering/pair distribution function beamline experiments (X-ray Free Electron Laser, Germany)
- Analysis of powder diffraction data (TOPAS, GudrunX, PDFgetX3)
- Presentation skills: Adobe Illustrator, Microsoft PowerPoint, CrystalMaker
- LaTeX (writing papers, theses, reports), Microsoft Word, Microsoft Excel

Other transferable skills: Planning, organisation, time management, teamwork, professionalism, commitment, logical thinking, problem-solving, taking the initiative, ability to conduct myself in a friendly and welcoming manner.

Schools attended: 19th BCA/CCG Intensive Teaching School in X-ray Structure Analysis (Durham, UK), 2025.

Additional research experience:

Wellcome Trust Centre for Human Genetics, Oxford (21–25 August 2017)

Skills: DNA extraction, dilutions and DNA quantification, polymerase-chain reaction, agarose gel electrophoresis.

SECONDARY EDUCATION

The Judd School (2016–2018)

Distinction 1 in Cambridge Pre-U Chemistry, A Levels in Maths (A*), Physics (A*), Biology (A*).
Year 12 Physics Prize, Year 13 Physics Prize.

Tonbridge Grammar School (2011–2016)

Free Standing Maths Qualification (FSMQ), Additional Mathematics (A)
IGCSE (9 A*): Biology, Chemistry, English Language, English Literature, French, History, Mathematics, Spanish, Physics.
GCSE Music (A*), GCSE Short course Religious Studies (A*).
IGCSE Chemistry Prize.

TEACHING EXPERIENCE

College Lecturer (Tutor) in Chemistry, St Catherine's College, University of Oxford (2024–2026)

Responsibilities: prepare teaching materials, give small-group classes (1st, 2nd, and 3rd Years), create tests, mark informal exams ('collections'), give feedback, create questions for and conduct undergraduate chemistry admission interviews, do safeguarding and unconscious bias training, attend college open days.

Mentor/supervise master's project students (2023–2026)

Responsibilities: supervise day-to-day experiments in the lab to teach practical techniques and ensure safety, discuss theory, teach coding, attend project meetings and offer ideas for future directions.

Private tutor in Chemistry, Maths, Physics, Biology for ages 10–18

Responsibilities: arrange, organise, and teach weekly sessions, create lesson plans and resources, give academic support and advice to students.

Volunteer classroom assistant, Chiddingstone Primary School (10 week from January 2016)

Responsibilities: supervise activities, read with students, give extension maths classes.

EXTRA-CURRICULAR INTERESTS

Cricket (Oxford University Women's Cricket Captain 2025, county cricket 2009–2022, House Sports Captain)

Dance (Ballroom, Latin American, Ballet, Modern, Jazz, Tap)

Music (Violin, Viola, Piano, Kent County Youth Orchestra 2015–2018, House Music Captain)

REFERENCES

Prof. Andrew Goodwin (Master's, PhD supervisor), Dept. Chemistry, University of Oxford, andrew.goodwin@chem.ox.ac.uk
Assoc. Prof. Ludmilla Steier (Tutor in Inorganic Chemistry), St Catherine's College, Univ. Oxford, ludmilla.steier@chem.ox.ac.uk
Prof. Luet Wong (Tutor in Inorganic Chemistry), St Hugh's College, University of Oxford, luet.wong@chem.ox.ac.uk